

Module 2: Transport Layer and Network Layer



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User Datagram Protocol (UDP)

Definition

The **User Datagram Protocol (UDP)** is a **connectionless** and **unreliable** transport-layer protocol.

Characteristics

- Provides process-to-process communication.
- Adds very little overhead.
- No connection establishment.
- No connection termination.
- Suitable for small and delay-sensitive messages.

Key Point

UDP provides only process-to-process communication on top of IP, without providing reliability, flow control, or congestion control.

UDP User Datagram Format

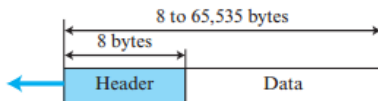
UDP Packet

A UDP packet is called a **User Datagram**.

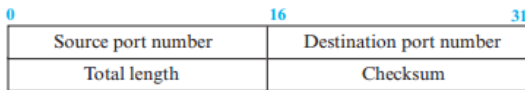
Header Characteristics

- Fixed header size: **8 Bytes**
- Four fields
- Each field occupies **16 bits (2 Bytes)**

Figure 3.39 *User datagram packet format*



a. UDP user datagram



b. Header format

UDP Header Fields

Field	Size	Purpose
Source Port	16 bits	Identifies sender process
Destination Port	16 bits	Identifies receiver process
Length	16 bits	Total length (Header + Data)
Checksum	16 bits	Optional error detection

Maximum UDP Datagram Size

65,535 Bytes

Exam Point

UDP data length equals

Total Length – 8 Bytes

since the UDP header is always 8 bytes.

Example 3.11: UDP Header Analysis

UDP Header

CB84 000D 001C 001C

Source Port	$\text{CB84}_{16} = 52100$
Destination Port	$\text{000D}_{16} = 13$
Total Length	$\text{001C}_{16} = 28 \text{ Bytes}$
Data Length	$28 - 8 = 20 \text{ Bytes}$
Direction	Client $\rightarrow$ Server
Server Process	Daytime Service

Observation

Destination port 13 is a **well-known port**, indicating that the packet is sent from a client to a server.

UDP Services

Services Provided by UDP

UDP offers only a subset of the general transport-layer services.

UDP Services

- Process-to-Process Communication
- Connectionless Service
- Optional Checksum
- Encapsulation and Decapsulation
- Queuing
- Multiplexing and Demultiplexing

Key Point

UDP is intentionally simple. It does not provide Flow Control, Error Control, or Congestion Control.

Process-to-Process Communication and Connectionless Service

Process-to-Process Communication

- UDP identifies communicating processes using
 - IP Address
 - Port Number
- Together they form a **Socket Address**.

Connectionless Service

- Each UDP datagram is independent.
- No connection establishment.
- No connection termination.
- Datagrams are not numbered.
- Different datagrams may follow different network paths.

Exam Point

A UDP message must fit into a single user datagram. The maximum application data size is

Flow Control and Error Control in UDP

Flow Control

- No Flow Control.
- No Sliding Window.
- Receiver may overflow.
- Application must handle flow control if required.

Error Control

- No retransmission.
- No acknowledgments.
- No duplicate detection.
- Corrupted packets are silently discarded.
- Optional checksum detects transmission errors.

Key Point

UDP provides only error detection through the checksum, not error recovery.

Checksum Calculation

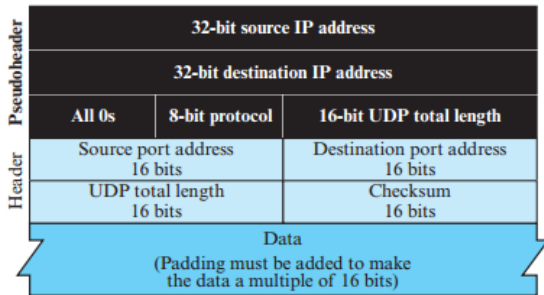
The UDP checksum is calculated using three components:

- ① IP Pseudoheader
- ② UDP Header
- ③ Application Data

Purpose of the Pseudoheader

- Detect incorrect destination host.
- Verify the protocol field (UDP = 17).
- Improve error detection reliability.

Figure 3.40 *Pseudoheader for checksum calculation*



Optional UDP Checksum (Example 3.12)

Checksum Rules

The sender may choose whether or not to calculate the UDP checksum.

Situation	Checksum Sent
Checksum not calculated	All 0s
Checksum calculated and result = All 1s	All 1s
Checksum calculated and result = All 0s	Impossible case

Important

A checksum value of all zeros indicates that the sender did not calculate the checksum.

Congestion Control and Encapsulation

Congestion Control

- UDP does not provide congestion control.
- Assumes messages are small and sporadic.
- Suitable for delay-sensitive applications.
- Congestion handling is left to the application if required.

Encapsulation and Decapsulation

- Sender encapsulates application data into a UDP datagram.
- Receiver decapsulates the UDP datagram and delivers the message to the application.

Key Point

UDP performs only basic transport-layer functions with minimum overhead.

Queuing, Multiplexing and Demultiplexing

Queuing

- UDP associates queues with ports.
- Client requests a port number from the operating system.
- Implementations may use:
 - Incoming queue only
 - Incoming and outgoing queues

Multiplexing and Demultiplexing

- Many application processes share a single UDP protocol.
- UDP multiplexes outgoing data from multiple processes.
- UDP demultiplexes incoming datagrams to the correct process using destination port numbers.

UDP vs Generic Simple Protocol

Generic Simple Protocol

- Connectionless
- No Flow Control
- No Error Control
- No Acknowledgment

UDP

- Connectionless
- No Flow Control
- No Retransmission
- Optional Checksum
- Corrupted packets discarded

Comparison

UDP is almost identical to the Generic Simple Protocol, except that UDP provides an optional checksum for detecting corrupted packets.

UDP Features

Major Features of UDP

The characteristics of UDP make it suitable for some applications and unsuitable for others.

Advantages

- Connectionless communication
- Low protocol overhead
- Fast transmission
- Low delay
- Supports multicasting

Limitations

- No reliability
- No flow control
- No congestion control
- No retransmission
- No packet ordering

Key Point

UDP is preferred when speed and low latency are more important than reliable delivery.

UDP Applications: Examples from the Textbook

Application	Reason
DNS	Short request and short response fit in one UDP datagram.
SMTP	Not suitable because long e-mails may require multiple datagrams that can arrive out of order.
Large File Download	Reliable delivery is required; UDP is not suitable.
Skype / Real-Time Audio & Video	Small packet loss is preferable to retransmission delay.

Examples

- Example 3.13 — DNS uses UDP.
- Example 3.14 — SMTP prefers TCP.
- Example 3.15 — File transfer requires reliable delivery.
- Example 3.16 — Real-time multimedia benefits from UDP.

Typical UDP Applications and Summary

Typical UDP Applications

- DNS (Domain Name System)
- TFTP (Trivial File Transfer Protocol)
- SNMP (Simple Network Management Protocol)
- RIP (Routing Information Protocol)
- Multicasting
- Interactive Real-Time Audio and Video

Key Takeaways

- UDP is a connectionless, unreliable transport-layer protocol.
- Uses an 8-byte header with four fields.
- Provides process-to-process communication using port numbers.
- Offers optional checksum for error detection.
- Does not provide flow control, error recovery, or congestion control.
- Best suited for lightweight request-response services and real-time applications where low delay is critical.